

Cohort Lens: OMOP All-Conditions Query

Diagnosis Feasibility Study Webapp

1. Cohort Conditions

This document contains the complete OMOP CDM query for the demonstration cohort.

T0 index: Essential hypertension (SNOMED 59621000)
Index window: 2020-01-01 through 2023-12-31
Age: 18 through 75 years at T0
Inclusion: Systolic blood pressure (LOINC 8480-6) >= 120 within 0-180 days
Inclusion: Metformin (RxNorm 861007) within 0-90 days
Exclusion: Hyperlipidemia (SNOMED 55822004) within -365 to 0 days

Sex is unrestricted (Any). T0 is the earliest matching index event for each patient.

2. Complete DuckDB Query

```
WITH PersonBase AS (  
  SELECT  
    person_id,  
    CASE  
      WHEN birth_datetime IS NOT NULL  
        THEN CAST(birth_datetime AS DATE)  
      WHEN year_of_birth IS NOT NULL  
        AND month_of_birth IS NOT NULL  
        AND day_of_birth IS NOT NULL  
        THEN MAKE_DATE(  
          CAST(year_of_birth AS INTEGER),  
          CAST(month_of_birth AS INTEGER),  
          CAST(day_of_birth AS INTEGER)  
        )  
      END AS birth_date  
  FROM person  
,  
  
  IndexEvents AS (  
    SELECT  
      co.person_id,  
      CAST(co.condition_start_date AS DATE) AS event_date  
    FROM condition_occurrence co  
    JOIN concept c  
      ON c.concept_id = co.condition_concept_id  
    WHERE c.concept_code = '59621000'  
      AND co.condition_start_date IS NOT NULL  
      AND CAST(co.condition_start_date AS DATE)  
        BETWEEN DATE '2020-01-01' AND DATE '2023-12-31'  
  ),  
  
  IndexCohort AS (  
    SELECT  
      person_id,  
      MIN(event_date) AS t0_date  
    FROM IndexEvents  
    GROUP BY person_id  
  ),  
  
  BasePatients AS (  
    SELECT  
      p.person_id,  
      i.t0_date  
    FROM PersonBase p  
    JOIN IndexCohort i  
      ON i.person_id = p.person_id  
    WHERE DATE_DIFF('year', p.birth_date, i.t0_date)  
      BETWEEN 18 AND 75  
  ),  
  
  LabInclusion AS (  
    SELECT DISTINCT  
      b.person_id  
    FROM BasePatients b  
    JOIN measurement m
```

```

        ON m.person_id = b.person_id
    JOIN concept c
        ON c.concept_id = m.measurement_concept_id
    WHERE c.concept_code = '8480-6'
        AND m.measurement_date IS NOT NULL
        AND m.value_as_number >= 120
        AND DATE_DIFF(
            'day',
            b.t0_date,
            CAST(m.measurement_date AS DATE)
        ) BETWEEN 0 AND 180
    ),

DrugInclusion AS (
    SELECT DISTINCT
        b.person_id
    FROM BasePatients b
    JOIN drug_exposure d
        ON d.person_id = b.person_id
    JOIN concept c
        ON c.concept_id = d.drug_concept_id
    WHERE c.concept_code = '861007'
        AND d.drug_exposure_start_date IS NOT NULL
        AND DATE_DIFF(
            'day',
            b.t0_date,
            CAST(d.drug_exposure_start_date AS DATE)
        ) BETWEEN 0 AND 90
    ),

ExcludedPatients AS (
    SELECT DISTINCT
        b.person_id
    FROM BasePatients b
    JOIN condition_occurrence co
        ON co.person_id = b.person_id
    JOIN concept c
        ON c.concept_id = co.condition_concept_id
    WHERE c.concept_code = '55822004'
        AND co.condition_start_date IS NOT NULL
        AND DATE_DIFF(
            'day',
            b.t0_date,
            CAST(co.condition_start_date AS DATE)
        ) BETWEEN -365 AND 0
    ),

IncludedCohort AS (
    SELECT b.person_id
    FROM BasePatients b
    JOIN LabInclusion l
        ON l.person_id = b.person_id
    JOIN DrugInclusion d
        ON d.person_id = b.person_id
    ),

FinalCohort AS (
    SELECT i.person_id
    FROM IncludedCohort i
    WHERE NOT EXISTS (
        SELECT 1
        FROM ExcludedPatients e
        WHERE e.person_id = i.person_id
    )
    )

SELECT
    (SELECT COUNT(*) FROM person) AS totalPatients,
    (SELECT COUNT(*) FROM IndexCohort) AS indexEligibleCount,
    (SELECT COUNT(*) FROM BasePatients) AS demographicCount,
    (SELECT COUNT(*) FROM IncludedCohort) AS inclusionCount,
    (SELECT COUNT(*) FROM FinalCohort) AS finalCount;

```

3. Expected Result

```

totalPatients      6731
indexEligibleCount 1351

```

demographicCount	1047
inclusionCount	28
finalCount	6

The query uses `person_id` to link events, `concept_code` to identify standard concepts, `DATE_DIFF` for timing relative to T0, and separate inclusion and exclusion CTEs for readable cohort logic.